

# Yujin Choi

Department of Computer Science and Engineering, Sejong University <https://uzn36.github.io/>

## RESEARCH POSITIONS

---

|                          |                                                                                                                                                                 |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sep. 2026<br>– Present   | <b>Assistant Professor</b> , Sejong University<br><i>Department of Computer Science and Engineering</i>                                                         |
| Apr. 2026<br>– Aug. 2026 | <b>Postdoctoral Researcher</b> , Nanyang Technological University, Singapore<br><i>The College of Computing and Data Science</i><br>Host: Prof. Jaehong Yoon    |
| Mar. 2026<br>– Aug. 2026 | <b>Postdoctoral Researcher</b> , Ulsan National Institute of Science and Technology<br><i>Department of Industrial Engineering</i><br>Mentor: Prof. Saerom Park |

## EDUCATION

---

|                          |                                                                                                                                                                                                                        |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mar. 2021<br>– Feb. 2026 | <b>Ph.D.</b> in Industrial Engineering, <b>Seoul National University</b> , Seoul, Korea<br>Dissertation: <i>Trustworthy Generative Models: from Privacy to Fairness</i><br>Advisor: Prof. Jaewook Lee, GPA: 3.84 / 4.3 |
| Mar. 2017<br>– Feb. 2021 | <b>B.Eng.</b> in Industrial Engineering & <b>B.Sc.</b> in Mathematics (Double Major)<br><b>Yonsei University</b> , Seoul, Korea<br>GPA: 3.91 / 4.3                                                                     |

## PUBLICATIONS

---

(†: equal contribution.; \*: corresponding author.; underline = 1<sup>st</sup>/corr. author.)

### Journals

- [J10] Y. Choi, J. Park, Y. Park, & J. Lee\*. Multiple Hyperplanes with Dual Encoding Support Vector Machine for Industrial Anomaly Detection. *IEEE Transactions on Industrial Informatics*, IF: 9.8, Accepted
- [J9] H. Kim, J. Park, W. Lee, J. Lee, & Y. Choi\*. Exploring the Effect of Multi-Step Ascent in Sharpness-Aware Minimization. *IEEE Access*. IF: 3.6, top 35.0%
- [J8] Y. Choi, J. Park, Y. Park, J. Lee, & J. Byun\*. Differentially Private Upsampling for Enhanced Anomaly Detection in Imbalanced Data. *Engineering Applications of Artificial Intelligence* (2026). IF: 8.0, top 2.8%
- [J7] Y. Choi, D. Kim, & J. Lee\*. Temporal Consistency Ensemble Empirical Mode Decomposition for Forecasting Practical Metal Price. *Engineering Applications of Artificial Intelligence*, 158, 111490 (2025). IF: 8.0, top 2.8%
- [J6] J. Byun, Y. Choi, & J. Lee\*. Improving the Utility of Differentially Private Clustering through Dynamical Processing. *Pattern Recognition*, 157, 110890 (2025). IF: 7.6, top 9.0%
- [J5] J. Byun, Y. Choi, J. Lee, & S. Park\*. Privacy-Preserving Inference Resistant to Model Extraction Attacks. *Expert Systems with Applications*, 256, 124830 (2024). IF: 7.5, top 6.6%
- [J4] J. Park, H. Kim, Y. Choi, W. Lee, & J. Lee\*. Fast Sharpness-Aware Training for Periodic Time Series Classification and Forecasting. *Applied Soft Computing*, 144, 110467 (2023). IF: 6.6, top 13.7%
- [J3] J. Byun, S. Park, Y. Choi, & J. Lee\*. Efficient Homomorphic Encryption Framework for Privacy-Preserving Regression. *Applied Intelligence*, 53(9), 10114–10129 (2023). IF: 3.5, top 41.2%
- [J2] Y. Choi, J. Park, J. Lee, & H. Kim\*. Exploring Diverse Feature Extractions for Adversarial Audio Detection. *IEEE Access*, 11, 2351–2360 (2023). IF: 3.6, top 35.0%
- [J1] J. Park, Y. Choi, J. Byun, J. Lee, & S. Park\*. Efficient Differentially Private Kernel Support Vector Classifier for Multi-Class Classification. *Information Sciences*, 619, 889–907 (2023). IF: 6.8, top 8.1%

### Top Conferences

- [C11] Y. Choi, & J. Yoon\*. Safe Few-Step Generation via Velocity Editing, Accepted, *NeurIPS 2026*
- [C10] D.Kim†, G.Shin†, Y. Choi, S.Park, & J.Lee\*. A Locally Tokenized Generative Model for Robust Time-Series Watermarking, Accepted, *NeurIPS 2026*

- [C9] W. Jeong†, **Y. Choi**†, D. Kim, S. Park & J. Lee\*. Leveraging Pathology Co-occurrence for Test-Time Adaptation in Chest X-Ray Diagnosis, *Medical Image Computing and Computer Assisted Intervention (MICCAI)* 2026, *accepted*
- [C8] H. Kim, J. Park, **Y. Choi**, & J. Lee\*. Stability Analysis of Sharpness-Aware Minimization. *International Conference on Machine Learning (ICML)* 2026, *accepted*
- [C7] J. Park, S. Lee, W. Jeong, **Y. Choi**, & J. Lee\*. Leveraging Priors via Diffusion Bridge for Time Series Generation. *SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)* 2026
- [C6] J. Park, **Y. Choi**, & J. Lee\*. Multi-Class Support Vector Machine with Differential Privacy. *Advances in Neural Information Processing Systems (NeurIPS)* 2025
- [C5] **Y. Choi**†, Y. Park†, J. Byun, J. Lee, & J. Park\*. Safeguarding Privacy of Retrieval Data against Membership Inference Attacks: Is This Query Too Close to Home?. *Findings of EMNLP 2025, Suzhou, China*
- [C4] J. Park†, **Y. Choi**†, & J. Lee\*. In-Distribution Public Data Synthesis with Diffusion Models for Differentially Private Image Classification. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 12236–12246 (2024)
- [C3] **Y. Choi**†, J. Park†, H. Kim, J. Lee, & S. Park\*. Fair Sampling in Diffusion Models through Switching Mechanism. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 38(20), 21995–22003 (2024)
- [C2] H. Kim, J. Park, **Y. Choi**, & J. Lee\*. Fantastic Robustness Measures: The Secrets of Robust Generalization. *Advances in Neural Information Processing Systems (NeurIPS)*, 36, 48793–48818 (2023)
- [C1] J. Park, H. Kim, **Y. Choi**, & J. Lee\*. Differentially Private Sharpness-Aware Training. *International Conference on Machine Learning (ICML)*, 27204–27224 (2023)

## PATENTS

---

|            |                                                                                                                                        |
|------------|----------------------------------------------------------------------------------------------------------------------------------------|
| 2025.05.28 | Clustering Method and Apparatus Supporting Differential Privacy<br>Reg. No. 10-2815746   App. No. 1020220185186, filed 2022.12.27      |
| 2023.06.07 | Apparatus and Method for Learning Differential Privacy Multi-Classification Based on Kernel Support Vector<br>App. No. 10-2023-0073178 |
| 2023.07.05 | Method and System for Detecting Adversarial Speech Example<br>App. No. 1020230087024                                                   |

## PREPRINTS & UNDER REVIEW

---

- [4] Y. Hee, **Y. Choi**, & J. Yoon\*. Confidence-Aware Tool Orchestration for Robust Video Understanding, Submitted, *ICLR* 2026
- [3] **Y. Choi**, J. Park, J. Byun, & J. Lee\*. Leveraging Programmatically Generated Synthetic Data for Differentially Private Diffusion Training, *1<sup>st</sup> major revision*, *Information Science* (IF: 6.8)
- [2] J. Park, **Y. Choi**, Y. Park, & J. Lee\*. Privacy-Preserving Asymmetric Banach Kernel Encoding for Support Vector Classification. *Submitted to Pattern Recognition* (IF: 7.6)
- [1] H. Kim, J. Park, **Y. Choi**, S. Lee, & J. Lee\*. BayesNAM: Leveraging Inconsistency for Reliable Explanations. *1<sup>st</sup> revision*, *Pattern Recognition* (IF: 7.6)

## SCHOLARSHIPS AND FELLOWSHIPS

---

|           |                                                                                                                               |
|-----------|-------------------------------------------------------------------------------------------------------------------------------|
| 2025      | Youlchon AI Scholarship, Youlchon AI Star                                                                                     |
| 2023–2025 | Doctoral Student Research Grant, Ministry of Science and ICT, Korea (Graduate Research Fellowship in Science and Engineering) |
| 2019–2020 | National Science and Technology Scholarship (full tuition), Ministry of Science, Korea                                        |

## TEACHING EXPERIENCE

---

### Lecturer

|             |                                                               |
|-------------|---------------------------------------------------------------|
| Spring 2025 | Financial Machine Learning, Soongsil University, Seoul, Korea |
| Fall 2024   | Financial Programming 2, Soongsil University, Seoul, Korea    |

Teaching Assistant (TA), Industrial Engineering (Prof. Jaewook Lee), SNU

2022 Mathematical Methods for Industrial Management, Seoul National University

**TA & Instructor, Corporate Education Programs, SNU**

2023–2024 HD Hyundai: Nonlinear Time Series Analysis

2022 Woori Bank: Big Data Analysis with Python and Deep Learning with PyTorch

2022–2024 SNU Big Data FinTech: Optimization and Machine Learning

2024–2025 SNU Big Data AI CEO Program

2021–2025 Samsung Data Scientist for Device Solution: Optimization

2021–2022 Samsung Data Scientist for Device Solution: Linear Algebra

**Samsung Best TA Award, 2021**

**REFERENCES**

---

**Jaewook Lee** Professor of Industrial Engineering, Seoul National University

[jaewook@snu.ac.kr](mailto:jaewook@snu.ac.kr)

**Saerom Park** Assistant professor of Industrial Engineering, UNIST

[srompark@unist.ac.kr](mailto:srompark@unist.ac.kr)

**Jaehong Yoon** Assistant professor of Computer and Data Science, Nanyang Technological University

[jaehong.yoon@ntu.edu.sg](mailto:jaehong.yoon@ntu.edu.sg)

**Hoki Kim** Assistant professor of Industrial Security, Chung-Ang University

[hokikim@cau.ac.kr](mailto:hokikim@cau.ac.kr)